

Logarithmic boundary depth and the conditioning of the discrete inverse conductance problem on hyperbolic lattices

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Abstract

We study the discrete inverse conductance (Calderón) problem—recovering the edge conductances of a resistor network from boundary voltage–current data—on layer-truncated hyperbolic $\{7, 3\}$ tilings, and compare it with square and triangular lattice disks of matched node count N . Using an exact formula for the sensitivity Jacobian of the Dirichlet-to-Neumann (DtN) map, we find that its condition number κ grows polynomially on the hyperbolic lattice ($\log_{10} \kappa = 3.89$ at $N = 847$, inside a preregistered prediction band), while on the flat lattices it reaches $10^{9.7}$ at $N = 317$ and exceeds the resolution of IEEE double precision by $N \approx 400$. The mechanism is geometric: the maximal distance from a node to the boundary grows as $O(\log N)$ on the hyperbolic lattice and as $O(\sqrt{N})$ on flat ones. Exact rank computations over two finite fields show that every tested network is algebraically identifiable; the difference is one of conditioning, not of identifiability. The advantage persists when the hyperbolic boundary is subsampled to the flat lattice’s probe count (on the identifiable subspace, $\log_{10} \kappa = 2.97$ versus 9.72 at $N \approx 316$; a post hoc metric) and for the Neumann-to-Dirichlet map measured by current-injection hardware. For the time-resolved RC version of the network, a preregistered finite-size prediction that the Dirichlet spectral gap is constant was refuted; we show instead, by domain monotonicity and the positive spectral gap of the nonamenable $\{7, 3\}$ graph, that the gap decreases monotonically to a strictly positive limit, so the RC relaxation time is bounded uniformly in N , whereas it grows linearly in N on flat lattices. A two-integrator protocol (SciPy and rusty-SUNDIALS CVODE) with analytic controls is specified for the RC model; the SciPy leg passes both controls. All data, code and a claim-by-claim verification ledger are released.

1 Introduction

Whether data measured on the boundary of a system determine its interior is a question that recurs across physics: in the bulk-boundary correspondence of topological phases [1–3], and in holographic duality, where boundary entanglement data encode bulk geometry [4, 5]. Its simplest classical instance is the inverse conductivity problem on a resistor network: given the map from boundary voltages to boundary currents, recover the interior conductances, and quantify how measurement noise is amplified in doing so.

For planar networks with boundary nodes on the outer face (*circular planar* networks), identifiability is characterised combinatorially: a network is recoverable from its response matrix if and only if it is critical [6–8]. Stability is a separate matter. Resistor-network approaches to electrical impedance tomography (EIT) show that reconstructions become unstable as networks are refined [9–11], consistent with logarithmic stability estimates for the continuum and discrete Calderón problems [12].

Hyperbolic lattices—regular tessellations of the hyperbolic plane—have been realised in circuit quantum electrodynamics [13] and on printed circuit boards [14], and are used as discrete

models of holography [15, 16]. Their defining metric property for the present purpose is that the boundary of a truncated tiling contains a fixed positive fraction of its nodes, so every node lies within $O(\log N)$ steps of the boundary. We ask whether this translates into a better-conditioned inverse problem, and answer quantitatively, with preregistered predictions, exact certificates where floating point is insufficient, and a time-domain formulation suited to a table-top measurement.

We did not find prior work stating conditioning results for inverse conductance problems on hyperbolic lattices, within targeted searches of the EIT, resistor-network and hyperbolic-lattice literatures; this is a statement about those searches, not a claim of priority.

2 Model and methods

2.1 Networks

Layer-truncated $\{7, 3\}$ tilings (heptagons, three per vertex) are generated in the Poincaré disk by reflecting the central heptagon across its edges, breadth-first by tile layer $L = 1, \dots, 6$ ($N = 35$ to 5887). Square and triangular lattice disks of radius R provide the flat comparison. Faces are constructed explicitly, and a node is a boundary node if and only if it is incident to the outer face; this is necessary because full-degree nodes can touch the outer face on both lattice families. Euler’s relation $V - E + F = 1$ is checked for every instance. All conductances are $g_e = 1$ unless stated.

2.2 Dirichlet-to-Neumann map and its sensitivity

With $L = \sum_e g_e (\delta_a - \delta_b)(\delta_a - \delta_b)^\top$ the weighted graph Laplacian and nodes partitioned into boundary (b) and interior (i), the DtN map is the Schur complement

$$\Lambda = L_{bb} - L_{bi} L_{ii}^{-1} L_{ib}. \quad (1)$$

Let H be the harmonic extension ($H_b = I$, $H_i = -L_{ii}^{-1} L_{ib}$), so $\Lambda = H^\top L H$. Since the harmonic extension minimises the Dirichlet energy for fixed boundary data, its variation does not contribute at first order, and

$$\frac{\partial \Lambda}{\partial g_e} = (h_a - h_b)(h_a - h_b)^\top, \quad e = (a, b), \quad (2)$$

where $h_v^\top$ is the row of H for node v . All edge sensitivities thus follow from one factorisation of L_{ii} ; eq. (2) agrees with central finite differences to 1.2×10^{-11} in our tests. The Jacobian J has one column per edge, containing the strictly upper-triangular entries of $\partial \Lambda / \partial g_e$ (the diagonal is determined by the zero row sums). We report $\kappa = \sigma_{\max}(J) / \sigma_{\min}(J)$, the factor by which relative noise in Λ can be amplified in a linearised reconstruction; a case is labelled numerically singular when $\sigma_{\min} \leq \varepsilon_{\text{mach}} \sigma_{\max}$.

2.3 Exact rank certificates

Floating-point rank decisions become unreliable precisely where κ is large. Because L has integer entries, for any prime p with $\det L_{ii} \not\equiv 0 \pmod{p}$ the harmonic extension and eq. (2) can be evaluated exactly in $\mathbb{F}_p$, and $\text{rank}_{\mathbb{Q}} J \geq \text{rank}_{\mathbb{F}_p}(J \bmod p)$. A full-rank result modulo p therefore certifies full rank over $\mathbb{Q}$; for rank-deficient cases we require agreement between $p = 2^{31} - 1$ and $p = 998244353$.

Table 1: Conditioning of the DtN sensitivity Jacobian, unit conductances, full boundary. $d_{\max}$: maximal graph distance to the boundary. SING.: smallest singular value at or below $\varepsilon_{\text{mach}} \sigma_{\max}$ in IEEE double precision (the true κ is not resolved there). The $\{7, 3\}$, $L = 4$ row is the preregistered confirmatory point.

Lattice	N	$ E $	$ \partial G $	$d_{\max}$	$\log_{10} \kappa$
$\{7, 3\}$, $L = 1$	35	42	28	1	1.39
$\{7, 3\}$, $L = 2$	112	140	77	3	2.46
$\{7, 3\}$, $L = 3$	315	399	203	5	3.27
$\{7, 3\}$, $L = 4$	847	1078	532	7	3.89
square, $R = 3$	29	44	20	2	1.73
triangular, $R = 3.2$	37	87	21	2	3.42
square, $R = 6$	113	200	44	5	5.07
triangular, $R = 6.4$	151	401	49	5	8.67
square, $R = 10$	317	592	76	9	9.72
triangular, $R = 10.8$	421	1176	84	10	SING.
square, $R = 16$	797	1528	124	15	SING.
triangular, $R = 17.2$	1069	3071	133	16	SING.

2.4 Neumann-to-Dirichlet map

Hardware that injects currents and reads voltages measures the Neumann-to-Dirichlet map Λ^+ (the Moore–Penrose inverse, the inverse on the zero-sum subspace), with sensitivity $\partial\Lambda^+/\partial g_e = -\Lambda^+(\partial\Lambda/\partial g_e)\Lambda^+$.

2.5 RC network

Attaching a capacitance C from every node to ground and imposing boundary voltages V_b gives

$$C \dot{V}_i = -(L_{ii}V_i + L_{ib}V_b), \quad (3)$$

whose steady state is the harmonic extension and whose steady boundary currents are ΛV_b . The slowest relaxation time is $\tau = C/\lambda_{\min}(L_{ii})$ and the stiffness ratio is $\lambda_{\max}(L_{ii})/\lambda_{\min}(L_{ii})$. Equation (3) is the time-resolved signal a physical network sampled by a microcontroller produces.

2.6 Integrators and controls

Equation (3) is integrated with two independent implementations of variable-order BDF: SciPy’s `solve_ivp` [17] and the CVODE solver of rusty-SUNDIALS [18], a Rust implementation of the SUNDIALS suite [19] exposed to Python through PyO3. Both use $\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$. Two controls gate every reported integration. **K1 (known answer)**: the solution $V_i(t) = e^{-L_{ii}t/C}(V_i(0) - V_\infty) + V_\infty$ is computed by matrix exponentiation and must be reproduced to 10^{-5} relative error (max norm) at $t \in \{0.25, 1, 4, 16\}\tau$. **K2 (cross-representation)**: the integrated steady-state boundary currents under $V_b = e_j$ must equal the Schur-complement column Λe_j to 10^{-6} . K2 ties the time-domain model to the static inverse problem, so the same code path serves both a solver benchmark and the measurement model of the experiment.

2.7 Preregistration

Predictions for the confirmatory $\{7, 3\}$, $L = 4$ point, and for the probe-matched, Neumann-to-Dirichlet and RC analyses, were fixed in dated, version-controlled documents before the corresponding computations were run, including refutation criteria and rival functional forms [20]. Deviations are reported where they occur.

Table 2: Probe-matched control and Neumann-to-Dirichlet (NtD) map. The $\{7, 3\}$ boundary is subsampled to the square disk’s probe count; unused boundary nodes carry zero current. Rank: exact, over $\mathbb{F}_p$ for two primes (agreeing). n_2 : unmeasured interior nodes of degree 2. σ_r/σ_{r+1} : float64 gap at the certified rank. κ_{id} : condition number on the identifiable subspace (a post hoc metric; see text). κ_{NtD} : full boundary.

Lattice	N	probes	$ E $	rank	n_2	σ_r/σ_{r+1}	$\log_{10} \kappa_{\text{id}}$	$\log_{10} \kappa_{\text{NtD}}$
$\{7, 3\}$, $L = 2$	112	44	140	117	23	1.5×10^{13}	2.70	1.99
square, $R = 6$	113	44	200	200	0	—	5.07	5.41
$\{7, 3\}$, $L = 3$	315	76	399	306	93	3.2×10^{12}	2.97	2.42
square, $R = 10$	317	76	592	592	0	—	9.72	10.08

Table 3: Dirichlet spectral gap $\lambda_{\min}(L_{ii})$ of the RC network ($C = 1$), relaxation time $\tau = C/\lambda_{\min}$ and stiffness ratio $\lambda_{\max}/\lambda_{\min}$. ^e: exploratory, not preregistered (sparse shift-invert eigensolver).

Lattice	N	$\lambda_{\min}$	τ	$\lambda_{\max}/\lambda_{\min}$
$\{7, 3\}$	35	1.0000	1.00	4.8
$\{7, 3\}$	112	0.3636	2.75	14.9
$\{7, 3\}$	315	0.2133	4.69	25.8
$\{7, 3\}$	847	0.1534	6.52	36.1
square	29	1.1716	0.85	5.8
square	113	0.2198	4.55	35.4
square	317	0.0688	14.54	115.3
square	797	0.0256	39.06	311.5
triangular	37	1.3258	0.75	6.3
triangular	151	0.2703	3.70	32.8
triangular	475	0.0768	13.02	116.7
triangular	1213	0.0281	35.56	319.6
$\{7, 3\}^{\text{e}}$	2240	0.1232	8.11	—
$\{7, 3\}^{\text{e}}$	5887	0.1058	9.45	—

3 Results

3.1 Conditioning versus network size

Table 1 and Fig. 1 give κ for all three families. On the $\{7, 3\}$ lattice the maximal boundary depth is $d_{\max} = 2L - 1$, and $\log_{10} \kappa$ rises from 1.39 ($N = 35$) to 3.89 ($N = 847$). The local exponent $d \log \kappa / d \log N$ *decreases* over the tested range (2.1, 1.8, 1.4), so κ grows no faster than N^2 there. On the square lattice the local exponent increases (5.7, then 10.4), $\log_{10} \kappa$ reaches 9.72 at $N = 317$, and from $N \approx 400$ onwards (triangular $N = 421$, square $N = 797$, triangular $N = 1069$) the Jacobian is numerically singular in double precision. At matched $N \approx 316$ the gap is 6.4 decades.

The preregistered confirmatory point, $\{7, 3\}$ at $L = 4$, was predicted to have $d_{\max} = 7$, $N \in [800, 900]$ and $\log_{10} \kappa = 4.2 \pm 0.5$; the measurements are 7, 847 and 3.89.

The mechanism is the scaling of depth with size. Boundary counts of the truncations, 28, 77, 203, 532, satisfy $b_{n+2} = 3b_{n+1} - b_n$, which grows geometrically (ratio $(3 + \sqrt{5})/2$), so the boundary is a fixed fraction of the nodes and $d_{\max} = O(\log N)$. On a flat disk $d_{\max} \sim R \sim \sqrt{N}$. In both geometries $\log \kappa$ increases with depth; only on the flat lattice does depth grow as a power of N .

Table 4: Integrator controls for the RC network ($\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$). K1: max-norm relative error against the matrix-exponential solution at $t \in \{0.25, 1, 4, 16\}\tau$, threshold 10^{-5} . K2: relative error of the integrated steady-state boundary currents against the Schur-complement column Λe_j , threshold 10^{-6} .

Network	Integrator	K1 error		K2 error	
$\{7, 3\}$, $L = 2$	SciPy BDF	7.3×10^{-9}	pass	4.2×10^{-13}	pass
square, $R = 6$	SciPy BDF	1.4×10^{-8}	pass	4.1×10^{-13}	pass
all	rusty-SUNDIALS CVODE (BDF)	not run: Python extension not built			

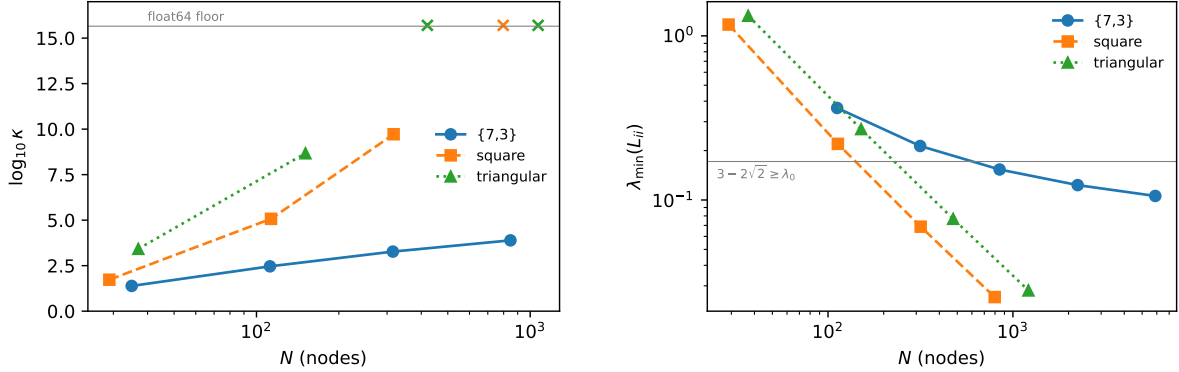


Figure 1: Left: $\log_{10} \kappa$ of the DtN sensitivity Jacobian versus N ; crosses mark numerically singular cases, plotted at the double-precision floor. Right: Dirichlet spectral gap $\lambda_{\min}(L_{ii})$ of the RC network; the $\{7, 3\}$ points at $N = 2240$ and 5887 are exploratory. The grey line is the upper bound $3 - 2\sqrt{2}$ on the bottom of the spectrum of the infinite $\{7, 3\}$ graph.

3.2 Identifiability is exact; the difference is conditioning

Exact ranks over $\mathbb{F}_p$ show that $\{7, 3\}$ at $L = 1, 2$, the square disk at $N = 113$, and triangular disks at $N = 187$ and $N = 475$ all have full column rank, certified for both primes. The last is comparable in size to flat instances whose floating-point SVD is at the precision floor. At the sizes tested, every edge is algebraically recoverable on both geometries; what separates them is the numerical precision needed to exploit that recoverability. This is consistent with the criticality theory of circular planar networks [6, 8], which concerns identifiability and is silent about conditioning.

3.3 Probe-matched control

A truncated hyperbolic tiling has more boundary probes than a flat disk of equal N (77 versus 44 at $N \approx 112$; 203 versus 76 at $N \approx 316$). To test whether the advantage is only a probe-count effect, the $\{7, 3\}$ boundary was subsampled to the flat probe count, evenly in angle, with unused boundary nodes carrying zero current.

The preregistered comparison of full-Jacobian condition numbers is undefined in this design: every subsampled Jacobian is singular. The exact ranks explain why. Each unused boundary node of degree two becomes an unmeasured interior node joining two conductances in series, of which only the series combination $g_1 g_2 / (g_1 + g_2)$ affects the boundary data. This is the standard non-recoverable series reduction of circular planar networks [7], and the exact deficiency equals the number of such nodes (23 at $L = 2$, 93 at $L = 3$; Table 2). The floating-point spectrum shows a gap of 10^{12} – 10^{13} at exactly the certified rank, so the float SVD resolves the rank unambiguously.

As a replacement metric, chosen after the singularity was observed and therefore post hoc, we report κ on the identifiable subspace, discarding exactly the certified number of singular values. It is $10^{2.70}$ versus $10^{5.07}$ at $N \approx 112$ and $10^{2.97}$ versus $10^{9.72}$ at $N \approx 316$. The two sides then recover different targets: 306 identifiable combinations of 399 hyperbolic conductances, versus all 592 square-lattice conductances. With that caveat, the conditioning advantage survives probe matching and is not a probe-count artefact at these sizes.

3.4 Neumann-to-Dirichlet map

For the map measured by current-injection hardware, the ordering is preserved, as preregistered: $\log_{10} \kappa_{\text{NtD}} = 1.99$ and 2.42 ($\{7, 3\}$, $L = 2, 3$) versus 5.41 and 10.08 (square, $R = 6, 10$) (Table 2).

3.5 RC dynamics and the Dirichlet spectral gap

The preregistered prediction was that $\lambda_{\min}(L_{ii})$ on $\{7, 3\}$ is constant over $L = 2-4$, refuted if it fell by a factor of two or more. It fell by 2.37 ($0.364 \rightarrow 0.153$), so the prediction is refuted as stated. The flat-lattice prediction, $\lambda_{\min} \sim 1/N$, holds (Table 3).

The asymptotic question is nevertheless settled by the following observation.

Proposition 1. *Suppose every interior node of a finite truncation G_L of the $\{7, 3\}$ tiling has degree 3. Then $\lambda_{\min}(L_{ii}(G_L)) \geq \lambda_0$, the bottom of the spectrum of the combinatorial Laplacian of the infinite $\{7, 3\}$ graph, and $\lambda_0 > 0$. If moreover the interiors are nested and exhaust the tiling, then $\lambda_{\min}(L_{ii}(G_L))$ is non-increasing in L and converges to λ_0 .*

Proof sketch. If interior nodes have full degree, then for f supported on the interior, $f^\top L_{ii} f = \sum_{\text{edges}} (f_a - f_b)^2$ with the sum over all edges of the infinite graph, so L_{ii} is the compression of the infinite Laplacian to the interior. By the Rayleigh quotient, $\lambda_{\min}(L_{ii})$ is the infimum of that quotient over the interior, which is at least λ_0 , the infimum over all finitely supported functions. Nesting gives monotonicity, and exhaustion gives convergence, since every finitely supported function is eventually supported in some interior. Finally $\lambda_0 > 0$ because the $\{7, 3\}$ graph has a positive edge-isoperimetric constant [21] and bounded degree [22, 23]. $\square$

For $L = 1, \dots, 6$ we verified by integer computation that every interior node has degree three and that the interior of G_L is exactly G_{L-1} (interior counts 7, 35, 112, 315, 847, 2240). For general L the hypothesis follows from the face-incidence definition of the boundary; this step is argued, not machine-checked. The computed values 0.364, 0.213, 0.153, 0.123, 0.106 ($L = 2-6$; the last two exploratory) decrease with successive ratios 0.59, 0.72, 0.80, 0.86, as the proposition requires, and lie below the tree bound $\lambda_0 \leq 3 - 2\sqrt{2} \approx 0.172$ from $L = 4$ on. The finite-size refutation and the uniform bound are compatible: the refuted prediction concerned the rate of approach to λ_0 , not its existence.

Physically, the slowest RC relaxation time of a hyperbolic network is bounded, $\tau \leq C/\lambda_0$, for every N , whereas on flat disks τ grows linearly in N . At $N \approx 800$, $\tau = 6.5C$ versus $39C$, and the stiffness ratio is 36 versus 311.

3.6 Integrator controls

Both controls pass for the SciPy integrator on $\{7, 3\}$, $L = 2$ and square $R = 6$, with K1 errors of order 10^{-8} and K2 errors of order 10^{-13} (Table 4). The rusty-SUNDIALS CVODE leg of the protocol has not been run for this version, because the Python extension was not built in the computing environment, and no cross-code agreement is claimed. The harness runs it automatically when the extension is importable, and the tables in this paper are generated from its output.

4 Discussion and limitations

The central result is geometric and simple: an inverse problem whose information decays with distance from the boundary is conditioned by the depth distribution of its unknowns, and hyperbolic truncations keep every unknown within $O(\log N)$ of the boundary. Negative curvature buys a better-conditioned discrete Calderón problem, and for the RC model a uniformly bounded relaxation time.

The limitations are as follows. (i) The flat-lattice growth law is not determined: three unsaturated flat data points do not distinguish $\kappa \sim e^{c\sqrt{N}}$ from a steep power law, and “numerically singular” states only that double precision cannot resolve κ , not its magnitude. (ii) κ is a linearised sensitivity at unit conductances. No nonlinear reconstruction algorithm is benchmarked, and the behaviour at strongly inhomogeneous conductances is not studied. (iii) Exact rank certificates cover $N \leq 475$ for flat lattices; full rank there does not exclude deficiency at larger N . (iv) The probe-matched comparison uses a post hoc metric and compares different identifiable targets. (v) Only $\{7, 3\}$ is studied among hyperbolic tilings. (vi) The CVODE cross-validation is specified but not yet executed. (vii) The novelty search was targeted, not systematic.

5 Research directions

Cross-validated RC benchmark with rusty-SUNDIALS. Equation (3) with the K1/K2 controls is a stiff linear system with an exact solution and a static cross-check, at tunable size and stiffness. Its stiffness is bounded in N on hyperbolic networks and grows linearly on flat ones. This makes it a natural regression benchmark for rusty-SUNDIALS, and running both integrators is the immediate next step. Supplying the analytic Jacobian $-L_{ii}/C$ to CVODE, which the current binding does not accept, would let solver cost be studied as a function of geometry.

Difference imaging. Localising a change δg of a few conductances from $\Lambda(g + \delta g) - \Lambda(g)$ is better conditioned than absolute recovery. Its depth dependence on hyperbolic versus flat networks is the quantity most directly relevant to a physical demonstration.

Other tilings and a scaling law. Repeating the analysis on $\{8, 3\}$, $\{5, 4\}$ and $\{p, q\}$ families would test whether the local exponent of κ is controlled by the boundary growth rate, i.e. by the curvature. Extended-precision or exact computation would extend the flat data beyond $N \approx 400$.

Tree metrics. Gromov-hyperbolic graphs are quasi-trees, and effective resistance on trees is additive, which links boundary resistance data to tree-metric and circular split-system reconstruction [24]. A stability theory of that reconstruction might explain the polynomial conditioning found here.

Tabletop measurement. A printed-circuit $\{7, 3\}$ network at $L = 2$ ($N = 112, 140$ resistors), with a matched flat control, can be probed in the time domain through eq. (3). The predicted observables are the relaxation time, the stiffness, and the conditioning of the measured Neumann-to-Dirichlet map. Since $\kappa \approx 10^{2.5}$ at this size, a linearised static reconstruction to 10% requires a total relative measurement error of about 3×10^{-4} . That precision budget, rather than component tolerance alone, determines whether the static comparison, or only the time-domain one, is feasible at $L = 2$.

Formalisation. The proposition above, and the finite-dimensional core of the bulk-boundary correspondence for the Su–Schrieffer–Heeger chain, are candidates for machine-checked proofs in Lean 4 with Mathlib, whose Schur-complement library already contains the block-matrix identities used here.

Verification status

Every claim in this paper is recorded in a machine-readable ledger [20] with an evidence tier: X (numerical), C (argument), L (literature), B (exact computation). The ledger’s consistency gate reports no findings.

Claim	Tier	Ledger
Conditioning versus N (Table 1), preregistered $L = 4$ point	X	H0-X-0001/0002
Full rank of all tested full-boundary networks	B	H0-B-0001
Probe-matched deficiency = number of unmeasured degree-2 nodes	B	H0-B-0002
Probe-matched κ on identifiable subspace (post hoc)	X	H0-X-0003
Neumann-to-Dirichlet ordering	X	H0-X-0004
H2 finite-size prediction refuted; flat $\lambda_{\min} \sim 1/N$	X	H2-X-0001
$\lambda_{\min}$ at $L = 5, 6$ (exploratory)	X	H2-X-0002
Interior degree 3, nested interiors, $L \leq 6$	B	H2-B-0001
$\lambda_0 > 0$ for $\{7, 3\}$	L	H2-L-0001
Uniform bound $\tau \leq C/\lambda_0$ (Proposition)	C	H2-C-0001
Integrator controls K1/K2 (SciPy); CVODE not run	X	H2-X-0003

Data and code availability

Code, preregistrations, raw data (JSON), network edge lists and the claim ledger are available at [20] and in the archived dataset accompanying this preprint. Every table in this paper is generated from the released data by `paper/make_assets.py`.

Use of AI tools

Code, numerical experiments and manuscript drafting were carried out with the assistance of Claude (Anthropic) as a research and programming assistant. The author reviewed all claims against their evidence and is responsible for the content.

References

- [1] C. L. Kane and E. J. Mele. Z_2 topological order and the quantum spin Hall effect. *Phys. Rev. Lett.*, 95:146802, 2005.
- [2] M. Z. Hasan and C. L. Kane. Colloquium: Topological insulators. *Rev. Mod. Phys.*, 82:3045, 2010.
- [3] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig. Classification of topological insulators and superconductors in three spatial dimensions. *Phys. Rev. B*, 78:195125, 2008.
- [4] J. Maldacena. The large N limit of superconformal field theories and supergravity. *Adv. Theor. Math. Phys.*, 2:231–252, 1998.
- [5] S. Ryu and T. Takayanagi. Holographic derivation of entanglement entropy from the anti-de Sitter space/conformal field theory correspondence. *Phys. Rev. Lett.*, 96:181602, 2006.

- [6] E. B. Curtis, D. Ingerman, and J. A. Morrow. Circular planar graphs and resistor networks. *Linear Algebra Appl.*, 283:115–150, 1998. doi: 10.1016/S0024-3795(98)10087-3.
- [7] E. B. Curtis and J. A. Morrow. *Inverse Problems for Electrical Networks*, volume 13 of *Series on Applied Mathematics*. World Scientific, Singapore, 2000.
- [8] Y. Colin de Verdière. Réseaux électriques planaires I. *Comment. Math. Helv.*, 69:351–374, 1994. doi: 10.1007/BF02564493.
- [9] L. Borcea, V. Druskin, F. Guevara Vasquez, and A. V. Mamonov. Resistor network approaches to electrical impedance tomography. In *Inside Out II*, volume 60 of *MSRI Publications*. Cambridge University Press, 2012.
- [10] L. Borcea, V. Druskin, A. V. Mamonov, and F. Guevara Vasquez. Circular resistor networks for electrical impedance tomography with partial boundary measurements. *Inverse Problems*, 26:045010, 2010. doi: 10.1088/0266-5611/26/4/045010.
- [11] L. Borcea, V. Druskin, A. V. Mamonov, and F. Guevara Vasquez. Pyramidal resistor networks for electrical impedance tomography with partial boundary measurements. *Inverse Problems*, 26:105009, 2010. doi: 10.1088/0266-5611/26/10/105009.
- [12] S. Ervedoza and F. de Gournay. Uniform stability estimates for the discrete Calderón problems. *Inverse Problems*, 27:125012, 2011.
- [13] A. J. Kollár, M. Fitzpatrick, and A. A. Houck. Hyperbolic lattices in circuit quantum electrodynamics. *Nature*, 571:45–50, 2019.
- [14] P. M. Lenggenhager et al. Simulating hyperbolic space on a circuit board. *Nat. Commun.*, 13:4373, 2022.
- [15] M. Asaduzzaman, S. Catterall, J. Hubisz, R. Nelson, and J. Unmuth-Yockey. Holography on tessellations of hyperbolic space. *Phys. Rev. D*, 102:034511, 2020.
- [16] P. Basteiro, G. Di Giulio, J. Erdmenger, J. Karl, R. Meyer, and Z.-Y. Xian. Towards explicit discrete holography: Aperiodic spin chains from hyperbolic tilings. *SciPost Phys.*, 13:103, 2022.
- [17] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods*, 17:261–272, 2020.
- [18] X. Callens. rusty-SUNDIALS: a Rust implementation of SUNDIALS solvers with Python bindings. Software, <https://github.com/xaviercallens/rusty-SUNDIALS>, commit af4886f, 2026.
- [19] A. C. Hindmarsh, P. N. Brown, K. E. Grant, S. L. Lee, R. Serban, D. E. Shumaker, and C. S. Woodward. SUNDIALS: Suite of nonlinear and differential/algebraic equation solvers. *ACM Trans. Math. Softw.*, 31(3):363–396, 2005.
- [20] X. Callens. Track H: hyperbolic resistor-network conditioning – code, data, preregistrations and claim ledger. <https://github.com/xaviercallens/SocrateAI-Scientific-CondensedMatterTheory>, directory experiments/track_h_hyperbolic_network, 2026.
- [21] O. Häggström, J. Jonasson, and R. Lyons. Explicit isoperimetric constants and phase transitions in the random-cluster model. *Ann. Probab.*, 30(1):443–473, 2002.
- [22] J. Dodziuk. Difference equations, isoperimetric inequality and transience of certain random walks. *Trans. Amer. Math. Soc.*, 284:787–794, 1984.

- [23] B. Mohar. Isoperimetric inequalities, growth, and the spectrum of graphs. *Linear Algebra Appl.*, 103:119–131, 1988. doi: 10.1016/0024-3795(88)90224-8.
- [24] S. Forcey and D. Scalzo. Phylogenetic networks as circuits with resistance distance. *Front. Genet.*, 11:586664, 2020.